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Hierarchical Algorithms
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Exercise sheet 06, Solution hints
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Homework:

Exercise 1: Proof the following statements:

a) Let $\alpha, c_2 > 0$, $c_1 \geq 0$ and let p be a polynomial. Then, there exist constants $\tilde{c}_1 \geq 0$ and $\tilde{c}_2 > 0$ with
$$p(k)c_1 \exp(-c_2 k^\alpha) \leq \tilde{c}_1 \exp(-\tilde{c}_2 k^\alpha) \quad \forall k \in \mathbb{N}_0.$$

b) Let $c_1, \tilde{c}_1 \geq 0$ and $c_2, \alpha > 0$ as well as $k \in \mathbb{N}_0$. For functions $R(x,y)$ and $\widetilde{R}(x,y)$, it holds
$$\|R\| \leq c_1 \exp(-c_2 k^\alpha) \quad \text{und} \quad \|\widetilde{R}\| \leq \tilde{c}_1 \exp(-c_2 k^\alpha).$$
Then, there exist $\hat{c}_1 \geq 0$ and $\hat{c}_2 > 0$ with
$$\|R + \widetilde{R}\| \leq \hat{c}_1 \exp(-\hat{c}_2 (2k)^\alpha).$$
What does this mean for the sum of exponentially convergent separable approximations?

c) Let $c_1, \tilde{c}_1 \geq 0$, $c_2, \alpha > 0$ and $k \in \mathbb{N}_0$. Let $\|\cdot\|$ be a submultiplicative norm. For $i \in \{1,2\}$ let $g_i(x,y)$ be functions with $\|g_i\| < \infty$ and let $g_i^s(x,y)$ corresponding separable expansions with remainders $R_i(x,y)$ and separation rank k, i.e. $g_i(x,y) = g_i^s(x,y) + R_i(x,y)$. For the remainders holds
$$\|R_1\| \leq c_1 \exp(-c_2 k^\alpha) \quad \text{und} \quad \|R_2\| \leq \tilde{c}_1 \exp(-c_2 k^\alpha).$$
Then, there exist $\hat{c}_1 \geq 0$ and $\hat{\alpha}, \hat{c}_2 > 0$ with
$$\|g_1 \cdot g_2 - g_1^s \cdot g_2^s\| \leq \hat{c}_1 \exp(-\hat{c}_2 (kk)^{\hat{\alpha}}).$$
What does this mean for the product of exponentially convergent separable approximations?

Solution hints:

a) For all $c > 0$, it holds $p(k) = \mathcal{O}(\exp(ck^\alpha))$. In particular for $c \in (0, c_2)$, there exist $\hat{c}_1 \geq 0$ and $k_0 \in \mathbb{N}_0$ with
$$p(k) \leq \hat{c}_1 \exp(ck^\alpha) \quad \forall k \geq k_0.$$
Hence, it exists a $c_1' \geq 0$ with
$$p(k) \leq c_1' \exp(ck^\alpha) \quad \forall k \in \mathbb{N}_0.$$
Multiplication with $c_1 \exp(-c_2 k^\alpha) \geq 0$ leads to
$$p(k)c_1 \exp(-c_2 k^\alpha) \leq c_1' c_1 \exp((c - c_2)k^\alpha) \quad \forall k \in \mathbb{N}_0.$$
With $\tilde{c}_1 := c_1' c_1 \geq 0$ and $\tilde{c}_2 := c_2 - c > 0$, we obtain the statement.

b) We have

$$\begin{aligned}\|R+\widetilde{R}\| &\leq \|R\| + \|\widetilde{R}\| \leq c_1 \exp(-c_2 k^\alpha) + \tilde{c}_1 \exp(-c_2 k^\alpha) \leq 2 \cdot \max\{c_1, \tilde{c}_1\} \exp(-c_2 k^\alpha) \\ &= 2 \cdot \max\{c_1, \tilde{c}_1\} \exp\left(-\left(\frac{1}{2}\right)^\alpha c_2 \, (2k)^\alpha\right).\end{aligned}$$

With $\hat{c}_1 := 2 \cdot \max\{c_1, \tilde{c}_1\} \geq 0$ and $\hat{c}_2 := \left(\frac{1}{2}\right)^\alpha c_2 > 0$, we obtain the statement. Hence, the sum of two exponentially convergent (each for the constants c_2 and α) (each for the same separation rank k) separable approximations is separable of rank $2k$ and exponentially convergent as well.

c) We define $m := \max\{\|g_1\|, \|g_2\|\}$. Then,

$$\begin{aligned}\|g_1 \cdot g_2 - g_1^s \cdot g_2^s\| &= \|g_1^s \cdot g_2^s + g_1^s \cdot R_2 + R_1 \cdot g_2^s + R_1 \cdot R_2 - g_1^s \cdot g_2^s\| \\ &\leq \|g_1^s \cdot R_2\| + \|R_1 \cdot g_2^s\| + \|R_1 \cdot R_2\| \\ &\leq \|g_1 \cdot R_2\| + \|R_1 \cdot R_2\| + \|R_1 \cdot g_2\| + \|R_1 \cdot R_2\| + \|R_1 \cdot R_2\| \\ &\leq \|g_1\| \cdot \|R_2\| + \|R_1\| \cdot \|g_2\| + 3\|R_1\| \cdot \|R_2\| \\ &\leq m(\|R_2\| + \|R_1\|) + 3\|R_1\| \cdot \|R_2\| \\ &\leq m\left(c_1 \exp(-c_2 k^\alpha) + \tilde{c}_1 \exp(-c_2 k^\alpha)\right) + 3c_1 \exp(-c_2 k^\alpha)\tilde{c}_1 \exp(-c_2 k^\alpha) \\ &\leq 2m \max\{c_1, \tilde{c}_1\} \exp(-c_2 k^\alpha) + 3c_1\tilde{c}_1 \exp(-2c_2 k^\alpha) \\ &\leq (2m \max\{c_1, \tilde{c}_1\} + 3c_1\tilde{c}_1) \exp(-c_2 (kk)^{\frac{\alpha}{2}})\end{aligned}$$

holds. With $\hat{c}_1 := 2m \max\{c_1, \tilde{c}_1\} + 3c_1\tilde{c}_1 \geq 0$, $\hat{c}_2 := c_2 > 0$ and $\hat{\alpha} := \frac{\alpha}{2} > 0$, we obtain the statement. Hence, the product of two exponentially convergent separable approximations (each with same constants c_2 and α and same separation rank k) is separable with rank k^2 and convergent as well (if the considered norm is submultiplicative).

Exercise 2: For $\nu = (\nu_1, \ldots, \nu_d) \in \mathbb{N}_0^d$ and $x = (x_1, \ldots, x_d) \in \mathbb{R}^d$, we define by using the multiindex notation

$$\nu! := \prod_{i=1}^d \nu_i!, \qquad |\nu| := \sum_{i=1}^d \nu_i, \qquad x^\nu := \prod_{i=1}^d x_i^{\nu_i} \qquad \text{und} \qquad \partial^\nu := \prod_{j=1}^d \left(\frac{\partial}{\partial x_j}\right)^{\nu_j}.$$

Let $f\colon \mathbb{R}^d \to \mathbb{R}$ be a function that is analytic in all variables and let $x_0 \in \mathbb{R}^d$ be an expansion point. Then,

$$f(x) = \sum_{\nu \in \mathbb{N}_0^d} \frac{(x - x_0)^\nu}{\nu!} \, \partial^\nu f(x_0)$$

is the Taylor series of f w.r.t. the expansion point x_0 and

$$T_p(x) = \sum_{|\nu| \leq p} \frac{(x - x_0)^\nu}{\nu!} \, \partial^\nu f(x_0)$$

is the finite Taylor sum $p \in \mathbb{N}$ of f w.r.t. the expansion point x_0. Let $g\colon \mathbb{R}^d \times \mathbb{R}^d \to \mathbb{R}, (x, y) \mapsto g(x, y)$ be a function that is analytic in x. Proof that the finite Taylor sum of degree p of g w.r.t. the component x w.r.t. the expansion point x_0 is a separable approximation and determine the separation rank.

Solution hints:

For $\nu = (\nu_1, \dots, \nu_d) \in \mathbb{N}_0^d$ and $x = (x_1, \dots, x_d) \in \mathbb{R}^d$, we define

$$\partial_x^\nu := \prod_{j=1}^d \left(\frac{\partial}{\partial x_j} \right)^{\nu_j}$$

for functions $g \colon \mathbb{R}^d \times \mathbb{R}^d \to \mathbb{R}, (x, y) \mapsto g(x, y)$. Then,

$$\sum_{|\nu| \le p} \frac{(x - x_0)^\nu}{\nu!} \, \partial_x^\nu g(x_0, y)$$

is the finite Taylor sum of degree p of g w.r.t. the first component x w.r.t. the expansion point x_0. With $a_\nu(x) := (x - x_0)^\nu$ and $b_\nu(y) := \frac{1}{\nu!} \, \partial_x^\nu g(x_0, y)$ for all $\nu = (\nu_1, \dots, \nu_d) \in \mathbb{N}_0^d$ with $|\nu| \le p$ follows that this is a separable approximation of g with separation rank $k = k(p, d) := \#\{\nu \in \mathbb{N}_0^d : |\nu| \le p\}$. It holds that

$$k(p, d) = \#\{\tilde{\nu} \in \mathbb{N}_0^{d+1} : |\tilde{\nu}| = p\} = \binom{p+d}{d} = \frac{(p+d)!}{d! \, p!} = \binom{p+d}{p}.$$

This is obtained by identifying $\tilde{\nu} \in \mathbb{N}_0^{d+1}$ with a string of length $p + d$ that contains a "+" sign in exactly d places. The positions of the "+" signs represent the value of the individual components $\tilde{\nu}_1, \dots, \tilde{\nu}_{d+1}$.

Exercises (in class):

Exercise 3: Implement the separable approximation using Chebyshev interpolation for the one dimensional case. Complete the parts marked by `/* EXERCISE */` of the functions

- `double lagrange_1d(int nu, int degree, const double *nodes, double x);`

  to evaluate the νth Lagrange polynomial of degree `degree` at the nodes `nodes` at the point x and

- ```
  prkmatrix interpolate_rkmatrix(int d, int rows, int cols,
                                 const double *nodes_x, const double *nodes_y,
                                 int degree, double (*test_function)
                                 (int d, const double *x, const double *y));
  ```

  to construct a $\mathcal{R}(k, \texttt{rows}, \texttt{cols})$ matrix with d-variate Chebshev interpolation of a test function w.r.t. the variable x. It holds $k = (\texttt{degree} + 1)^d$ where `degree` is the degree of interpolation and the nodes are denoted by `nodes_x` and `nodes_y`. The parameter `rows` denotes the number of nodes in `nodes_x` and the parameter `cols` denotes the number of nodes in `nodes_y`.

in *interpolation.c*.
Based on these methods, the methods for the multidimensional case are implemented already such as
  ```

```
double lagrange_md(int d, const int *nu, int degree, const double *nodes,
                   const double *x);
```

for the evaluation of the $(\nu_1, \ldots, \nu_d)$th d-variate Lagrange polynomial of degree `degree` for the nodes `nodes` at the position x.

In *example_ 06.c*, the newly implemented routines should then be used to calculate a rank-k approximation for a matrix defined by evaluating a kernel function. For this, n points $(x_i)_{i=1}^n$ and $(y_i)_{i=1}^n$ in $\mathbb{R}^d$ are generated, and the matrix to be approximated is defined as

$$A := (g(x_i, y_i))_{1 \le i,j \le n} \in \mathbb{R}^{n,n} \quad \text{with} \quad g : \mathbb{R}^d \times \mathbb{R}^d, g(x, y) := \exp(-\|x - y\|_2^2).$$

In the following, R_K^{poly} denotes the rank-k approximation to A calculated by interpolation and R_K^{best} denotes the best rank-k approximation to A calculated by SVD.

In *example_ 06.c*, adjust the choice of nodes $(x_i)_{i=1}^n$ and $(y_i)_{i=1}^n$ such that:

- Test 1: $(x_i)_{i=1}^n$, $(y_i)_{i=1}^n \subset [0, 1]^d$,
- Test 2: $(x_i)_{i=1}^n \subset [0, 1]^d$, $(y_i)_{i=1}^n \subset [1, 4]^d$,
- Test 3: $(x_i)_{i=1}^n \subset [1, 4]^d$, $(y_i)_{i=1}^n \subset [0, 1]^d$.

Perform numerical tests for these three test examples to calculate the respective approximation errors and complete the following table. The call is made by explicitly specifying the number of nodes, the polynomial degree and the spatial dimension, i.e. in the form

```
./example_06 n degree dim
```

Explain the results.

d spatial dimension	n matrix size	degree polynomial degree	k rank	$\|A - R_k^{\text{poly}}\|_2$ interpolation approx. error			$\|A - R_k^{\text{best}}\|_2$ best-approx.error		
				Test 1	Test 2	Test 3	Test 1	Test 2	Test 3
1	1000	1	2	63.4	17.6	139.5	8.81	2.62	2.62
1	1000	2	3	8.71	4.50	38.3	0.38	0.20	0.20
1	1000	4	5	0.25	0.10	10.5	3.0e-4	6.9e-4	6.9e-4
1	500	10	11	3.8e-07	1.1e-07	6.1e-3	1.6e-13	2.2e-12	2.2e-12
1	1000	10	11	7.5e-07	2.3e-07	1.2e-2	3.1e-13	4.4e-12	4.4e-12
1	1500	10	11	1.1e-06	3.5e-07	1.9e-2	4.6e-13	6.4e-12	6.4e-12
2	1000	1	4	83.4	5.28	58.9	7.58	0.74	0.74
2	1000	2	9	7.70	1.69	15.1	0.32	5.1e-2	5.1e-2
2	1000	4	25	0.22	3.3e-2	4.24	3.5e-05	1.7e-05	1.7e-05
2	500	10	121	3.3e-07	3.6e-08	2.2e-3	1.8e-15	3.3e-16	3.3e-16
2	1000	10	121	6.7e-07	6.6e-08	4.6e-2	2.9e-15	1.0e-15	1.0e-15
2	1500	10	121	9.8e-07	1.2e-07	6.8e-2	3.9e-15	1.6e-15	1.6e-15
3	1000	1	8	99.2	1.80	20.0	6.44	0.13	0.13
3	1000	2	27	6.66	0.46	5.17	4.1e-2	1.0e-3	1.0e-3
3	1000	4	125	0.19	8.2e-3	1.22	3.5e-07	2.2e-08	2.2e-08
3	500	10	1331	2.9e-07	9.7e-09	7.9e-4	0	0	0
3	1000	10	1331	5.9e-07	1.9e-08	1.4e-3	0	0	0
3	1500	10	1331	8.7e-07	4.1e-08	2.4e-3	4.2e-16	1.8e-22	1.6e-22

Solution hints: